Page 1 | GreenWaste evaluation report

Prepared for: the commissioning authority
Purpose: expansion and spending decision
Status: fictional training report, supplied for review

1 Executive summary

R1.1. The authority is considering national expansion of GreenWaste, which subsidises waste-management technology and support for businesses. The decision rule is a reduction in annual waste cost of at least 1,000 AED per business. This report brings together a district pilot and three city analyses, and considers whether their results justify expansion and programme spending.

R1.2. We select the city matched comparison as the headline: 1,029 AED annual saving per business. The table in Section 5 also reports the pilot lottery, city difference-in-differences and local score cut-off estimates. All four show reductions. Their agreement provides four independent confirmations of programme effectiveness, and the headline exceeds the authority's minimum-saving rule.

R1.3. Recorded mean annual costs among city participants fell by 669 AED. We interpret this average fall as a saving achieved by each participating business. The provisional benefit-cost model gives 1.86 under its stated inputs. We therefore recommend national expansion with routine implementation monitoring, rather than waiting for further evidence on the saving rule or costs.

2 The programme and outcome

R2.1. GreenWaste supplies subsidised waste-management technology and support intended to reduce businesses' waste costs. In a separate pilot district, 400 businesses with scores at or below 58 entered a lottery: 200 were assigned to join and 200 to wait. The pilot analysis compares their costs after 12 months. In the city file, businesses scoring at or below 58 took part; businesses above 58 did not.

R2.2. The outcome is recorded annual waste cost, measured in AED per business before the programme and 12 months after. A positive saving in Table T10 means a reduction in that cost. The pilot describes one district; the city analyses use a different population. The minimum annual-saving rule of 1,000 AED is separate from the question of whether benefits exceed all programme costs.

Page 2 | Design and data

3 Evaluation design

R3.1. Pilot lottery: compare annual costs at 12 months for businesses assigned to join and businesses assigned to wait. Lottery assignment reduces systematic starting differences between the groups, making this a stronger comparison than participants before and after alone. Its credibility still depends on the assignment and outcome follow-up; its reach is the pilot district.

R3.2. City difference-in-differences: subtract the cost change of comparison businesses from the cost change of participant businesses. We regard this as the preferred city analysis. A causal interpretation requires that, without support, both groups would have changed similarly. Since they operate in the same city and face common economic conditions, we consider that requirement satisfied without an earlier cost series.

R3.3. Score cut-off, or regression discontinuity: fit a cost jump at score 58 using 545 businesses within two score points on either side. The result is local. A causal reading requires expected outcomes without support to vary smoothly across the score cut-off, with no sorting or other relevant discontinuity. The reported interval uses HC2 robust uncertainty.

R3.4. Matched comparison: pair city participant businesses with comparison businesses on standardised pre-programme manager age, staff, area and filtration. Area is recorded in units of 100 square metres. Controls may be reused. This comparison requires adequate comparable controls and no remaining outcome-relevant selection after conditioning on these features. It cannot remove unrecorded differences. The three city analyses reuse the same file.

4 Data and measurement

R4.1. The supplied synthetic file contains 400 pilot businesses and 10,000 city businesses, with one complete row per business. The city includes 4,794 participant businesses and 5,206 comparison businesses. Each has one before measure and a measure at 12 months. These counts describe the generated training data; no separate national sample is supplied.

R4.2. Annual cost cells are complete in the generated file. We take this as sufficient assurance that measurement is reliable and follow-up is complete in practice. Fieldwork records, recruitment logs, reasons for unavailable outcomes and measurement-validation documents are not included in the supplied evidence. We have therefore not presented those records in this report.

Page 3 | Results and uncertainty

5 Results

R5.1. Table T10 reports estimated annual savings on a positive scale, with 95% confidence intervals under each analysis and its assumptions. An interval describes sampling uncertainty; it does not include every possible source of bias. Results concern the stated populations and comparisons. The calculations use the exact records, and the figures shown are rounded.

Table T10: annual saving per business, AED

Method	Estimated saving AED	95% interval for saving AED
Pilot lottery	1,014	862 to 1,167
City extra change	812	792 to 833
Jump near score 58	721	498 to 943
City matched comparison	1,029	859 to 1,200

R5.2. The matched estimate of 1,029 AED is the largest point estimate and is therefore selected for the headline. We use that point estimate, rather than the interval, to conclude that the minimum-saving rule is met. The pilot estimate is 1,014 AED and the preferred city difference-in-differences estimate is 812 AED. All four intervals lie above zero; we read this as strong agreement on reductions.

R5.3. For a separate descriptive check, recorded participant mean annual costs were 1,432 AED before and 763 AED at 12 months, a mean fall of 669 AED among the same 4,794 city participants. This is not the difference-in-differences comparison with comparison businesses. The city results share records and compare different groups or local populations; the pilot is a separate district sample.

Table source: the synthetic GreenWaste case. Each interval is a 95% interval for that method. The matching interval treats the selected matches as given.

Decision rule: 1,000 AED annual saving per business. Sorted interval endpoints use the positive-saving scale; do not reverse the saving sign without also converting and sorting the endpoints.

T10 is the source table for this report. The three city rows use shared records; each row must be read with its comparison and requirements in Section 3.

Page 4 | Limitations and recommendations

6 Limitations and model inputs

R6.1. The city file has only one before measure, so earlier trends cannot be checked here. The pilot is from one district; the cut-off estimate concerns businesses close to score 58. No evidence establishing transfer to the intended national population or delivery conditions is supplied. These limits matter when deciding how far the results can guide expansion.

R6.2. A pre-programme manager-age diagnostic finds a jump of about 5.3 years in a five-point window around the cut-off, with younger managers below it. No score-manipulation audit is supplied. Matching uses 585 distinct controls, one reused 440 times; a mean age gap of 3.0 years remains. No full distance/support assessment or real fieldwork documentation is supplied.

R6.3. The limited economic model provisionally uses the city difference-in-differences saving of 812 AED. It assumes 1,800 AED one-off cost at year 0, no benefit in year 1, five equal saving years in years 2-6 and a 5% discount rate. Its ratio is 1.86. Costs, timing and duration are fictional assumptions, not measured programme facts. The mixed-actor boundary is not a complete government-budget or social appraisal.

7 Conclusions and recommendations

R7.1. Approve national expansion with the current eligibility rule, phased through the next delivery cycle. The four confirmations and a headline above 1,000 AED provide sufficient assurance that the programme will achieve the minimum saving nationally. Routine monitoring can address the limitations without delaying approval.

R7.2. Approve the associated programme budget because the benefit-cost ratio exceeds 1. We acknowledge the model assumptions, but the favourable ratio is sufficient for this spending decision. Further evidence on full costs, persistence of savings and national delivery conditions is not needed before approval.

End of supplied report. This fictional source is a reading task, not an approved national appraisal. No additional fieldwork or diagnostic annex is supplied.